Art Insider, an Augmented Reality Tool for Art Recognition and User Knowledge Enrichment.

Osvaldo E. Cascante
Maynooth University, Ireland
osvaldo.cascanteaguilar.2026@mumail.ie

***Abstract*—Art appreciation in museums, galleries, and cultural heritage sites is often limited by the lack of contextual knowledge of visitors, including the historical background, artistic techniques, and the history of their creator. This paper presents Art Insider, a mobile Augmented Reality (AR) application designed to enhance user engagement and understanding by delivering real time information about cultural and heritage content. The system addresses the challenge of recognizing different art forms by implementing a dual-pipeline architecture that combines deep learning based 2D recognition for paintings with model-based 3D recognition for sculptures and artifacts. For painting recognition, a custom dataset was created using Roboflow, incorporating variations in viewpoint, lighting, and occlusion. A YOLOv8n object detection model was trained and deployed in Unity via ONNX format using the Unity Inference Engine. For 3D objects, a separate pipeline utilizes photogrammetry generated with Scaniverse and Blender to create 3D meshes, which are then processed using Vuforia's Model Target Generator (MTG) to allow 360 ° recognition and tracking. Preliminary results demonstrate effective real-time performance, with accurate painting detection and stable 3D object tracking under varying viewpoints. Future work includes the full integration of both pipelines, enhanced user interaction through distance-aware interfaces, and the incorporation of multimedia content to further enrich the visitor experience.**

***Index Terms*—Cultural Heritage, Augmented Reality, 2D Object recognition, 3D Object recognition, Unity, YOLOv8n, Vuforia.**

I. Introduction

Augmented Reality technology allows users to perceive the world in an enhanced way; it improves users' experience by overlaying computer-generated, including graphics, sounds, and sometimes touch feedback, on the real environment [1]. Lack of knowledge and understanding of the arts can prevent many museum, gallery and cultural site visitors from fully appreciating a work of art, including its techniques, its creator's life and the historical context that shapes its meaning. By using AR technologies, galleries, libraries, archives, and museums (GLAMs) can add multiple sensorial experiences that can engage visitors with Cultural Heritage (CH) [2]. Cultural heritage includes all the tangible and intangible elements of a society inherited from previous generations and preserved for future generations [3]. Lately, Augmented Reality has enabled Cultural Heritage environments to create the missing spark from its static environment [4]. AR applications in cultural heritage environments have evolved from simple multimedia guides into sophisticated interactive systems. Existing solutions, such as early museum guide applications, typically rely on marker-based tracking. However, the development of SLAM (Simultaneous Localization and Mapping) has enabled Markerless Augmented Reality (MAR) tracking in museum settings. An example is 'MuseumEye,' described by Hammady et al. 2018, which integrates a combination of multimedia content, including audio commentaries, video presentations, layered textual information, and image galleries [5]. 2D artworks, such as paintings, are suited to deep learning-based object detection (OD). In contrast, 3D objects (sculptures and artifacts), require geometry aware recognition techniques that can handle viewpoint variation and spatial tracking. To address this challenge, this paper presents Art Insider, a mobile AR application that integrates a dual-pipeline recognition system for artworks. The first pipeline employs a deep learning model based on YOLOv8 for real-time detection and recognition of paintings, enabling the overlay of artwork contextual information. The second pipeline utilizes photogrammetry and model-based tracking through Vuforia Engine. Both pipelines are implemented within a unified Unity environment, enabling real-time performance on mobile devices. The main contributions of our work are:

- A dual-pipeline AR system for recognizing both 2D and 3D artworks in real time on a mobile device.
- An integrated Unity-based implementation that combines deep neural network learning inference through YOLOv8 and model-based tracking using Vuforia.

II. Relate Work and Literature Review

A. Augmented Reality in Cultural Heritage

Ramtohul et al, 2024 completed a systematic review of AR applications in the context of cultural heritage sites. They reviewed 45 works published between 2013-2023 [4]. They realized more and more cultural organizations are harnessing the potential of AR to enhance the quality of user experience. They highlighted the museum environment as well-suited for AR integration, reason why the number of AR projects in this sector is increasing. Ipalakava et al, 2025, presented the development of a mobile application for the Abylkhan Kasteyev State Museum of Arts (Almaty, Kazakhstan), designed to recognize and visualize exhibits through AR [6]. They combined computer vision and machine learning to identify artifacts using a smartphone. A study conducted at leading art museums in United States showed that almost all participants agreed that mobile technology enhanced their experiences; 9 out of 10 said it made it easier to access information; and 91% thought it was an exciting way to learn [7]. The National Gallery in London is another museum that has used augmented reality to take its collections beyond its walls with an experience that members of the public can access through their phones [8].

B. DataSets

Computer vision systems based on machine learning rely heavily on large and well-annotated datasets for training and evaluation. Due to technological improvements a large digitalization effort has been made, leading to an increasing availability of large digitalized visual art collections, e.g. WikiArt [9]. In addition, major museums have made their digitalized collections publicly available, for example, the Rijksmuseum through The Rijksmuseum Challenge [10] or The Metropolitan Museum of Art of New York City. Alternately, the vast amount of images available in internet enable researchers to create their own dataset using a variety of labeling tools like Label IMG, VGG Annotator, Label Studio or Roboflow [11].

C. Deep Learning Neural Networks for 2D Artwork Recognition

Machine learning today is one of the most important, and fastest growing, fields of technology. Applications of machine learning are becoming ubiquitous, and solutions learned from data are increasingly displacing traditional hand-crafted algorithms. One particular branch of machine learning, known as deep learning, has emerged as an exceptionally powerful and general-purpose framework for learning from data [12]. Convolutional Neural Network (CNN) are a type of Artificial Neural Network specifically designed to process images and visual data. CNN have revolutionized image processing and are now almost universally used in computer vision applications [13]. When used in the artistic domain, a CNN can learn to recognize an artist's visually distinctive features by adapting its filters to the presence of these features in a digitalized paining [9]. Many models have been developed over the years and have represented the state of the art, but for the interest of our research we will focus on YOLO (You Only Look Once). Redmon et all, 2016 described YOLO as a single convolutional network simultaneously predicts multiple bounding boxes and class probabilities for those boxes [14]. Since its appearance in 2016 YOLO has evolve into multiple versions of different sizes and capabilities from YOLOv1 to YOLOv11. In 2023, YOLOv8 was introduced offering five models with different sizes (e.g., nano, small, medium) that balance performance and computational cost. The "s" and "n" versions are fast and compact, suited for mobile apps with lower performance [15]. YOLOv8 is actively supported, with multiple training resources and is widely used in research projects.

D. 3D Object Recognition and Model-Based Tracking in AR

Photogrammetry is a widely used technique in cultural heritage for generating accurate 3D models from multiple overlapping 2D images. This approach has been increasingly adopted in museums for digital preservation and documentation. The resulting 3D meshes can be integrated into augmented reality systems. Apollonio et al. 2021 covers in great detail the pipeline from acquisition to visualization, to achieve a highly portable and accurate description in the shape and the appearance of 3D models inferred from museum's objects [16]. Photogrammetry is considered the best technique for the processing of image data, being able to deliver at any scale of application accurate, metric and detailed 3D information [17].

Model-based tracking in augmented reality is a markerless technique that uses a known 3D model of an object or environment to estimate a camera's pose (position and orientation). Unlike traditional marker-based AR, this approach relies on natural features like edges and textures of physical objects [18]. Syed et al, 2023 presented an in depth review of Augmented reality tracking technologies and development tools [19]. ARCore (Google-Android), ARKit(Aple-iOS) and Vuforia (PTC Inc.-iOS-Android-Windows) are three popular AR platforms used to create AR experiences. Vuforia includes additional features such as 3D object detection with 360° views recognition from 3D models. It employs advanced computer vision techniques to detect and track marks by analysing the camera's target features. It can recognize complex 3D objects [20]. It is not stated explicitly by Vuforia how their tracking algorithm's function but based on their documentation they utilize a variation of Natural Feature Tracking (NFT). This technique was introduced in the early 1980s and revolves around splitting an image into several sections [21]. It is a markerless approach in augmented reality that identifies and tracks environmental features—such as texture, contrast, and unique visual patterns—rather than artificial markers [22],[23].

III. Methodology & System Architecture

A. 2D Recognition Pipeline used for Paintings

We decided to use a custom dataset as our main objective during the creation of Art Insider is to learn and experiment with different tools rather than using a consolidated dataset like ArtWiki or the The Rijksmuseum Challenge. For this reason, we selected Roboflow as the labeling tool and built a Dataset of 10 masters paintings. Targeted to acquired 20 different images per painting from the web; in some cases, this was not feasible. In total we populated Roboflow with 178 images. We used three data augmentation techniques available within the free version of Roboflow to reach 428 images. We used 375 images as training set, 36 as validation set and 17 for test set. The object detection model was implemented using YOLOv8n (nano) variant, which is designed for efficient real-time performance on resource-constrained devices such as phones or tablets. The model was trained using the Ultralytics framework (version 8.4.21) in a CPU-based environment (11th Gen Intel Core i5-1135G7). Training was conducted for 80 epochs with an input image resolution of 640×640 pixels and a batch size of 8. The initial learning rate was set to 0.003, with a final learning rate factor of 0.01, and early stopping was applied with a patience of 15 epochs to prevent overfitting. The model was exported as an Open Neural Network Exchange

(ONNX) format. ONNX is a free and open-source framework that provides a standard format for describing AI models, allowing for interoperability and portability across tools and platforms [24]. We incorporate the YOLOv8n ONNX model into Unity's Inference Engine (formally known as Barracuda and Sentis). Camera Frame (s) are passed to Unity Inference Engine, and a custom script decodes the output tensor and displays a bounding box together with the highest probability painting name on screen. We explored a distance-driven user interface feature. If user is interested and move forward to the passersby distance, then the user can view more information [25]. As a single painting becomes the main object in the camera frame the application will display painting contextual information beyond the painting name.

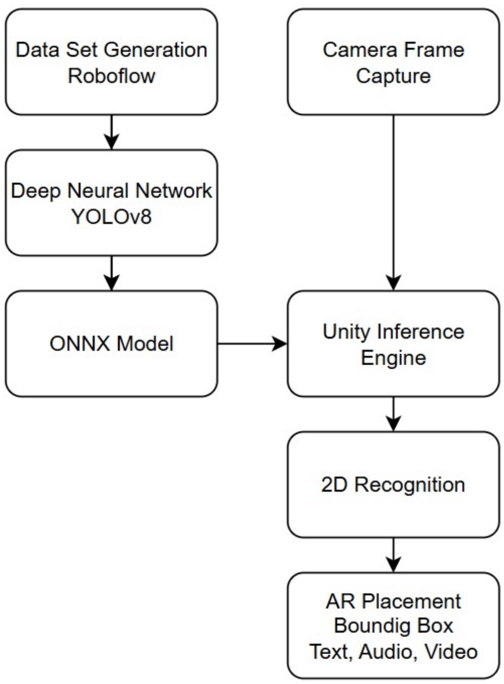

Fig. 1. Art Insider 2D Painting Recognition Flow.

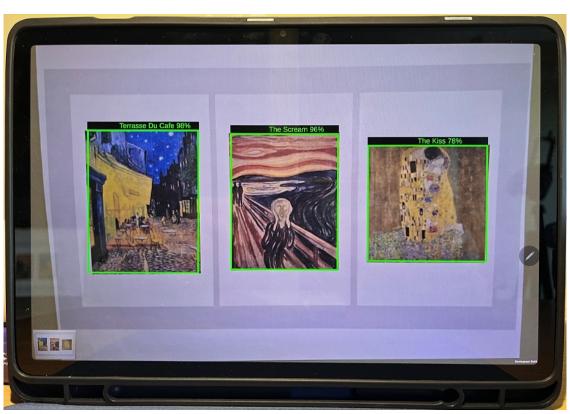

Fig. 2. Multiple paintings recognition, bounding box, painting name and YOLOv8n confidence level.

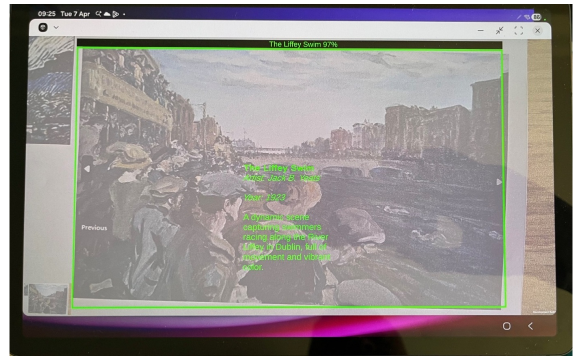

Fig. 3. Single painting detection with bounding box, painting name, YOLOv8n confidence level and contextual information overlayed.

B. 3D Recognition Pipeline: Sculptures and Artifacts

We used Scaniverse, a photogrammetry tool to generate 3D models (meshes). Scaniverse is a free, high-performance 3D scanning app for iOS and Android that specializes in creating "Digital Twins" of objects and environments. Scaniverse models often contain hundreds of thousands of polygons, for that reason we used Blender to reduce the density of the mesh while preserving the silhouette of the object. Then we upload the reduced 3D Mesh to Vuforia™ Model Targets Generator (MTG). For complex objects, the MTG uses Advanced Model Targets, which utilize AI to allow the app to recognize the object from any angle without requiring a specific Guide View. Advanced Model Targets support recognizing and tracking one or more objects from all sides, all contained in a single Advanced Model Target Database. Recognition ranges can be defined up to 360 degrees around every object without requiring the user to align an outline of the model with the physical object to start tracking [26]. Then we exported the generated database files Vuforia Engine for Unity to handle the 3D recognition.

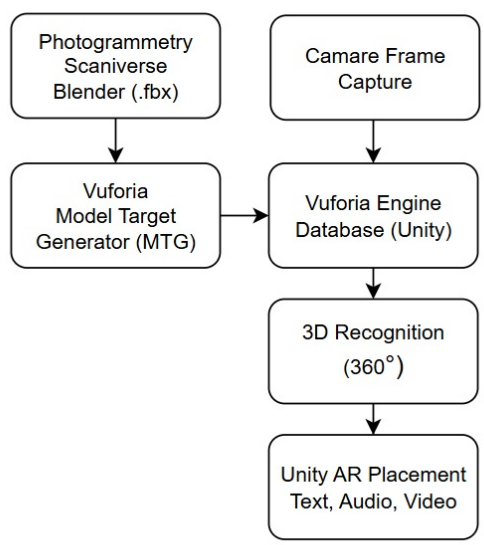

Fig. 4. Art Insider 3D Object Recognition Flow

IV. Results and Evaluation

A. 2D YOLO Model Results

The YOLOv8 model achieved on the validation set a precision of 0.965, Recall of 1.000, Mean Average Precision (mAP@0.5) of 0.995 and mAP@0.5:0.95 of 0.962. The model correctly identified nearly all instances of paintings in the validation dataset and produced accurate bounding boxes. The processing times per image was 84.1mS. Which means 11.8 images can be processed per second or approximately 12 frames per second (FPS) on CPU, sufficient for real-time applications. We must point out that the training, and validation sets are small (375 and 36 images). Not sufficient for a real application but enough for a proof-of-concept demonstration.

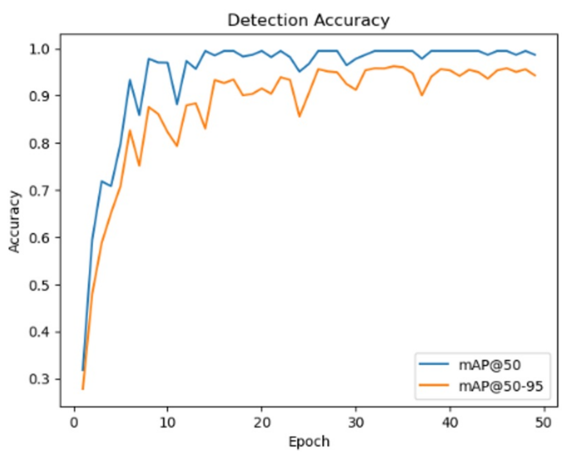

Fig. 5. 2D YOLOv8n Model detection accuracy

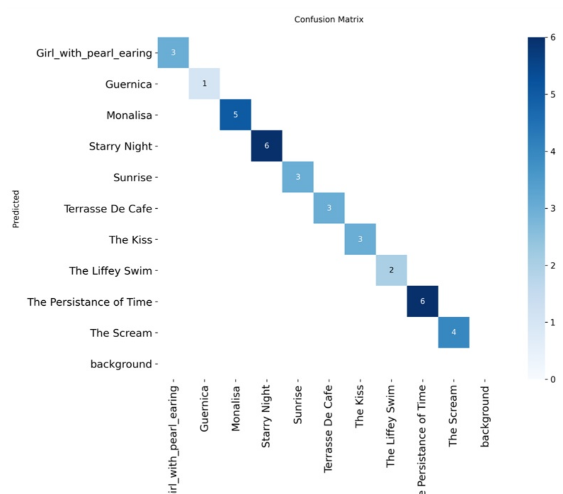

Fig. 6. 2D YOLOv8n Model confusion matrix

B. 2D Unity Inference Engine Result

We tested 5 sets of two paintings. The Code in Unity is using a detection threshold of 50% at 5 FPS to prevent overloading the GPU with inference cycles. We repeated this experiment 40 times for each set of paintings at 30cm camera to image distance to estimate system detection success rates. The system achieved an overall detection success rate of 70% across the 10 paintings. Paintings with high contrast, clear edges and bright colors achieved 100% detection success rate while dark painting with no edges nor bright colors achieved very low detection rates. Inference time on 70% of the paintings is perceived as real time. With one painting taking 0.5-1S and two painting failing detection.

TABLE I
2D Art Insider Detection Performance on Selected Paintings

Painting	Inference Time	Confidence	Success Rate
The Kiss	0.5S	< 70%	100%
The Persistence of Memory	Real-time	98%	100%
The Starry Night	Real-time	90%	100%
Café Terrace at Night	Real-time	95%	100%
Girl with a Pearl Earring	Real-time	< 55%	100% (unstable)
Mona Lisa	Failure	< 50%	0%
Impression, Sunrise	Failure	< 50%	0%
Guernica	Real-time	98%	100%
The Liffey Swim	Real-time	90%	100%
The Scream	Real-time	98%	100%

C. 3D Vuforia Engine for Unity

The performance of the 3D object recognition system was evaluated using one 3D model from different views and 3 elevation angles (10°, 40° & 80°) at a distance of 30cm. The system achieved a detection success rate of 98% across 200 trials, with a recognition time of less than 1 second. Tracking stability remained high, with the object successfully tracked 100% of the interaction time.

Fig. 7. 3D 10° elevation object front view recognition.

Fig. 8. 3D 40° elevation object front view recognition.

Fig. 9. 3D 80° elevation object front view recognition.

TABLE II
3D OBJECT DETECTION PERFORMANCE RESULTS

Vista	10° View	40° View	80° View
Front	100%	100%	100%
Left	94%	100%	98%
Right	100%	100%	100%
Back	100%	100%	100%

These results indicate that the system is suitable for real-time augmented reality applications involving cultural heritage objects.

V. LIMITATIONS AND FUTURE WORK

For the 2D recognition pipeline, we used a limited dataset of 10 paintings with 178 images augmented to 428 and a validation set of 36 images, which may lead to overly optimistic YOLOv8 performance metrics. This is not sufficient for an accurate model, but only for a proof of concept (POC). As shown in the Results section, the recognition rate is low at 70%, impacted by paintings of low contrast with no edges or bright colors. Both the number of paintings and number of images should be increased and should include museum real scenarios of lighting, reflections and occlusion. The use of a consolidated art dataset ie. ArtWiki should be a future avenue for research. In the 3D recognition pipeline, our study only used one 3D model. The sample size should be increased, and experimentation with real museum sculptures and artifacts should be implemented. As discussed by Ramtohul et al, 2024 the incorporation of Collaboration, Social interaction and Gamification [4] can bring the tool to the state of the art.

VI. CONCLUSION

This paper presented Art Insider, a dual-pipeline augmented reality system for real-time recognition of both 2D and 3D artworks on mobile devices. By integrating YOLOv8n based object detection for paintings with Vuforia's model-based tracking for 3D objects within Unity, the system demonstrates the possibility of combining deep learning and tracking techniques into an AR application. Experimental results showed that the system performs well on paintings with high contrast and edges, achieving real-time inference in most cases. However, overall performance reached 70% detection success rate, with reduced accuracy on low-contrast and no edges paintings. This work is intended as a proof of concept, and several limitations remain. The 2D recognition pipeline relies on a relatively small dataset of 10 paintings (178 images augmented to 428), which may lead to optimistic performance and limited generalization. Future work should expand both the number of paintings and image samples, incorporating real-world museum conditions such as varying lighting, reflections, and occlusions. The use of a larger, consolidated datasets such as ArtWiki would further improve model robustness. Similarly, the 3D recognition pipeline was evaluated using only a single model; future research should include a broader range of real sculptures and artifacts to assess scalability. Beyond technical improvements, future developments could explore the integration of collaborative features, social interaction, and gamification elements, as suggested in prior work, to enhance user engagement and move the system closer to state-of-the-art AR experiences in cultural heritage scenarios.

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